

Equilibrium Analysis of Beams: Support Reactions & Combined Loads

High-Level Overview

In structural engineering, determining **support reactions** is the foundational step in analyzing any loaded beam. When external forces, distributed loads (UDLs/UVLs), inclined point loads, and pure moment couples act on a beam, the supports must exert equal and opposite reactions to maintain **static equilibrium**.

To evaluate complex beam systems:

- Support Conditions** dictate the types and directions of unknown reaction forces:
 - Hinge (Pin) Support:** Exerts horizontal (H_A) and vertical (V_A) reaction forces.
 - Roller Support on Inclined Plane:** Exerts a single normal reaction force R perpendicular to the inclined plane angle α . The reaction vector thus acts at an angle $\theta = 90^\circ - \alpha$ relative to the beam line.
- Distributed Loads** must be converted into single equivalent concentrated forces acting at their specific centroids.
- Equilibrium Equations** ($\Sigma F_x = 0$, $\Sigma F_y = 0$, and $\Sigma M = 0$) are applied systematically to solve for all unknown reactions.

Analytical Breakdown of Support & Load Handling

1. Roller Support on an Inclined Surface

When a roller support rests on a plane inclined at an angle α to the horizontal:

- The reaction force vector R acts perpendicular to the inclined plane.
- Angle of reaction vector relative to horizontal: $\theta = 90^\circ - \alpha$.
- Resolution into components:
 - Horizontal component: $R_x = R \cos(90^\circ - \alpha)$
 - Vertical component: $R_y = R \sin(90^\circ - \alpha)$

For example, if the inclined plane angle is $\alpha = 30^\circ$, the normal reaction vector makes an angle of 60° ($90^\circ - 30^\circ$) relative to the horizontal beam axis:

- $R_x = R \cos 60^\circ$
- $R_y = R \sin 60^\circ$

2. Resolving Trapezoidal Load Distributions

A trapezoidal load varying from intensity w_1 to w_2 across span length L can be split into two basic geometry components:

$$\text{Trapezoidal Load} = \text{Rectangular UDL } (w_1 \times L) + \text{Triangular UVL } \left(\frac{1}{2} \times (w_2 - w_1) \times L \right)$$

- Rectangular Load Component (W_1):**

$$W_1 = w_1 \times L \quad \text{acting at distance } \frac{L}{2} \text{ from the start of the load}$$

- Triangular Load Component (W_2):**

$$W_2 = \frac{1}{2} \times (w_2 - w_1) \times L \quad \text{acting at distance } \frac{1}{3}L \text{ from the peak side (or } \frac{2}{3}L \text{ from the start)}$$

Worked Examples

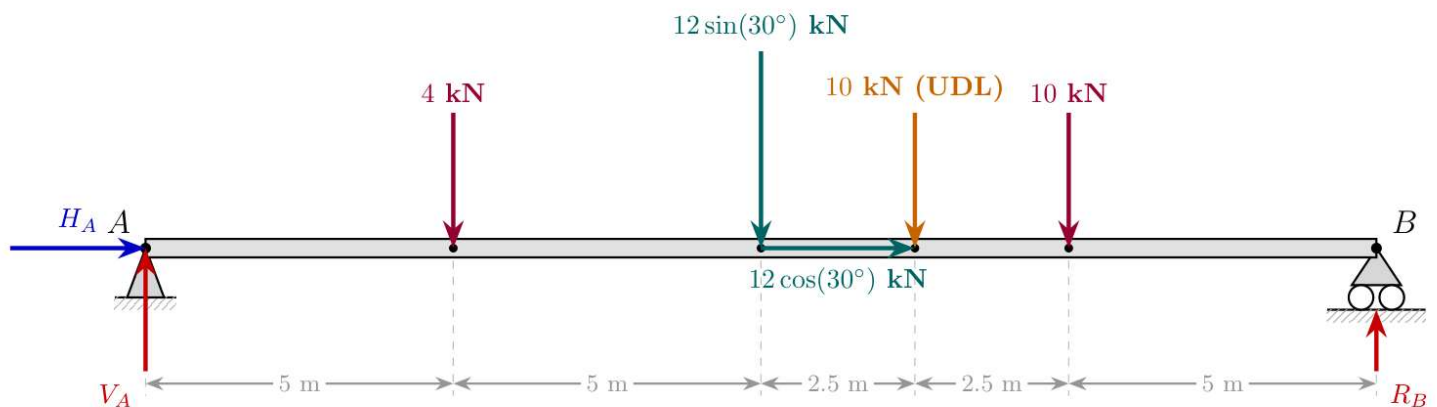
Problem 1: Simply Supported Beam with Point Loads, Inclined Force, and UDL

Problem Statement

A beam AB is hinged at end A and roller supported at end B . It is subjected to the following loads:

1. A vertical point load of 4 kN downward at 5 m from A .
2. An inclined force of 12 kN acting downward-rightward at 30° to the horizontal, applied at 10 m from A .
3. A UDL of intensity 2 kN/m spanning 5 m (between 10 m and 15 m from A).
4. A vertical point load of 10 kN downward at 15 m from A .
5. Total beam length between A and $B = 20$ m.

Find the support reactions at A and B .



Visual Description: Free body diagram of horizontal beam AB (20 m long). Hinge support at A with horizontal reaction H_A and vertical reaction V_A . Roller support at B with upward reaction R_B . Loads applied from left to right: 4 kN downward force at 5 m; inclined load at 10 m resolved into horizontal $12 \cos 30^\circ$ and vertical $12 \sin 30^\circ$; equivalent UDL force 10 kN downward at 12.5 m; vertical point load 10 kN downward at 15 m.

Step-by-Step Solution

Step 1: Force Resolution & Equivalent Loads

- **Inclined Force Components (12 kN at 30°):**
 - Horizontal component: $12 \cos 30^\circ \approx 10.392$ kN ($\rightarrow$)
 - Vertical component: $12 \sin 30^\circ = 6$ kN ($\downarrow$)
- **UDL Equivalent Point Load (2 kN/m over 5 m):**
 - Magnitude: $W_{UDL} = 2 \times 5 = 10$ kN
 - Location from A : $10 \text{ m} + 2.5 \text{ m} = 12.5 \text{ m}$

Step 2: Apply Horizontal Equilibrium ($\Sigma F_x = 0$)

$$H_A + 12 \cos 30^\circ = 0 \implies H_A + 10.392 = 0$$

$$\mathbf{H_A = -10.392 \text{ kN} \implies \mathbf{H_A = 10.392 \text{ kN} } (\leftarrow)$$

Step 3: Apply Moment Equilibrium About A ($\Sigma M_A = 0$)

Taking counter-clockwise moments as positive (+) and clockwise moments as negative (-):

$$\Sigma M_A = -(4 \times 5) - (12 \sin 30^\circ \times 10) - (10 \times 12.5) - (10 \times 15) + (R_B \times 20) = 0$$

$$-20 - 60 - 125 - 150 + 20R_B = 0$$

$$-355 + 20R_B = 0 \implies 20R_B = 355$$

$$R_B = 17.75 \text{ kN} \quad (\uparrow)$$

Step 4: Apply Vertical Equilibrium ($\Sigma F_y = 0$)

$$V_A - 4 - 12 \sin 30^\circ - 10_{\text{UDL}} - 10 + R_B = 0$$

$$V_A - 4 - 6 - 10 - 10 + 17.75 = 0$$

$$V_A - 30 + 17.75 = 0$$

$$V_A = 12.25 \text{ kN} \quad (\uparrow)$$

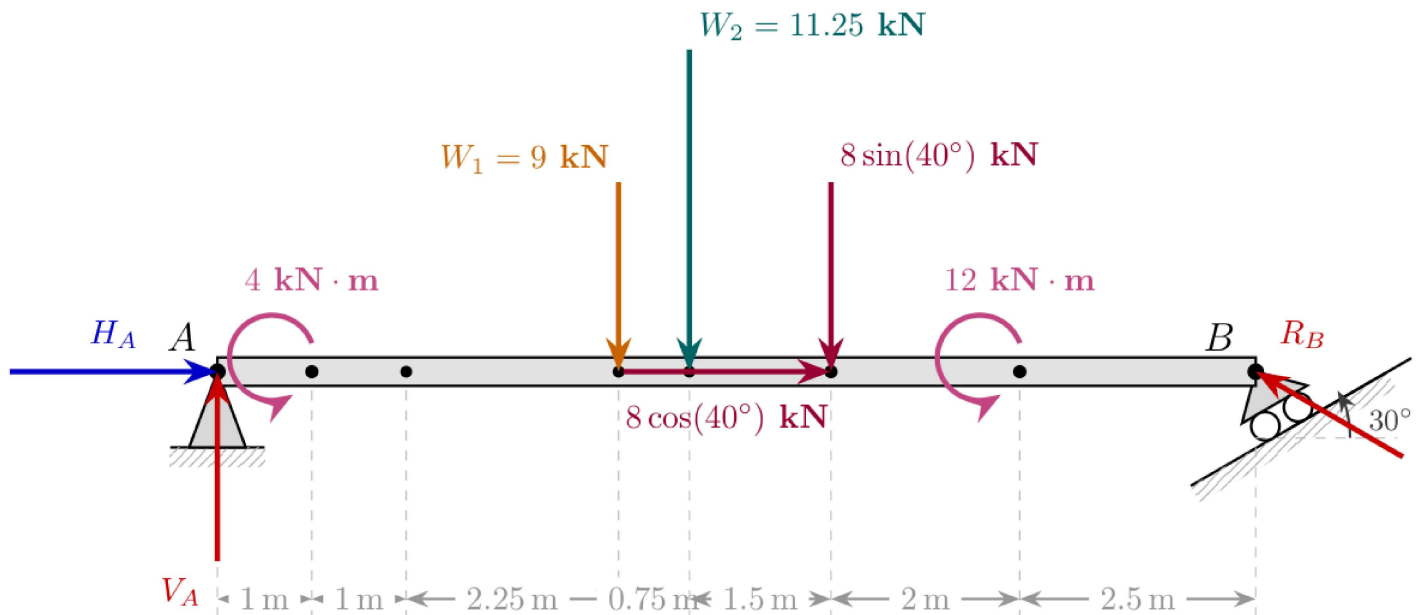
Problem 2: Beam with Trapezoidal Load, Inclined Roller Support, and Moment Couples

Problem Statement

A beam AB is hinged at A and supported by rollers on a 30° inclined surface at end B . The beam is subjected to:

1. A counter-clockwise couple moment of $4 \text{ kN} \cdot \text{m}$ at 1 m from A .
2. A trapezoidal distributed load starting at 2 m from A and spanning 4.5 m , with intensity increasing from 2 kN/m to 7 kN/m .
3. An inclined force of 8 kN at 40° to the horizontal directed downward-leftward, applied at the end of the trapezoidal load (6.5 m from A).
4. A counter-clockwise couple moment of $12 \text{ kN} \cdot \text{m}$ at 8.5 m from A .
5. Beam length from A to $B = 11 \text{ m}$.

Find the support reactions at A and B .



Visual Description: Beam AB (11 m span). Point A is hinged with reactions H_A and V_A . Point B is on a 30° inclined surface, giving normal reaction R at 60° to the beam axis. From A to B : $4 \text{ kN} \cdot \text{m}$ CCW couple at 1 m ; trapezoidal load (span 2 m to 6.5 m) split into 9 kN rectangular

load at 4.25 m and 11.25 kN triangular load at 3.5 m from A ; 8 kN inclined force at 6.5 m resolved into $8 \cos 40^\circ$ (leftward) and $8 \sin 40^\circ$ (downward); 12 kN · m CCW couple at 8.5 m; roller reaction R at B resolved into $R \cos 60^\circ$ (leftward) and $R \sin 60^\circ$ (upward).

Step-by-Step Solution

Step 1: Decomposition of Loads & Reaction Components

1. Trapezoidal Load Breakdown (Span $L = 4.5$ m, from 2 m to 6.5 m):

- Base intensity $w_1 = 2$ kN/m, peak intensity $w_2 = 7$ kN/m.
- Rectangular UDL Component (W_1):**

$$W_1 = w_1 \times L = 2 \times 4.5 = 9 \text{ kN} \quad (\downarrow)$$

Location from A : $2 \text{ m} + \frac{4.5 \text{ m}}{2} = 4.25 \text{ m}$

- Triangular UVL Component (W_2):**

$$W_2 = \frac{1}{2} \times (w_2 - w_1) \times L = \frac{1}{2} \times (7 - 2) \times 4.5 = 11.25 \text{ kN} \quad (\downarrow)$$

Location from A : $2 \text{ m} + \frac{1}{3}(4.5 \text{ m}) = 3.5 \text{ m}$ (since peak 7 kN/m is at left side, centroid is $1/3$ span from left)

2. Inclined Load (8 kN at 40°):

- Horizontal component: $8 \cos 40^\circ \approx 6.128$ kN ($\leftarrow$)
- Vertical component: $8 \sin 40^\circ \approx 5.142$ kN ($\downarrow$)

3. Inclined Roller Reaction at B (R at 60° to beam axis):

- Horizontal component: $R \cos 60^\circ$ ($\leftarrow$)
- Vertical component: $R \sin 60^\circ$ ($\uparrow$)

Step 2: Moment Equilibrium About A ($\Sigma M_A = 0$)

Taking CCW moments as positive (+) and CW as negative (−):

- Pure CCW couple moment at 1 m: $+4 \text{ kN} \cdot \text{m}$
- Triangular UVL load (11.25 kN at 3.5 m): $-(11.25 \times 3.5) = -39.375 \text{ kN} \cdot \text{m}$
- Rectangular UDL load (9 kN at 4.25 m): $-(9 \times 4.25) = -38.25 \text{ kN} \cdot \text{m}$
- Vertical component of 8 kN force ($8 \sin 40^\circ$ at 6.5 m): $-(8 \sin 40^\circ \times 6.5) \approx -33.425 \text{ kN} \cdot \text{m}$
- Pure CCW couple moment at 8.5 m: $+12 \text{ kN} \cdot \text{m}$
- Vertical component of roller reaction R at B (11 m): $+(R \sin 60^\circ \times 11)$

Setting the sum of moments to zero:

$$+4 - (11.25 \times 3.5) - (9 \times 4.25) - (8 \sin 40^\circ \times 6.5) + 12 + (R \sin 60^\circ \times 11) = 0$$

$$4 - 39.375 - 38.25 - 33.425 + 12 + 11R \sin 60^\circ = 0$$

$$-95.05 + 9.5263R = 0 \implies 9.5263R = 103.05$$

$$\mathbf{R = 10.82 \text{ kN}}$$

Step 3: Horizontal Force Equilibrium ($\Sigma F_x = 0$)

$$H_A - 8 \cos 40^\circ - R \cos 60^\circ = 0$$

$$H_A - (8 \times 0.7660) - (10.82 \times 0.5) = 0$$

$$H_A - 6.128 - 5.41 = 0 \implies H_A - 11.538 = 0$$

$$\mathbf{H_A = 11.538\text{ kN} \quad (\rightarrow)}$$

Step 4: Vertical Force Equilibrium ($\Sigma F_y = 0$)

$$V_A - 11.25 - 9 - 8 \sin 40^\circ + R \sin 60^\circ = 0$$

$$V_A - 11.25 - 9 - 5.142 + (10.82 \times 0.8660) = 0$$

$$V_A - 25.392 + 9.370 = 0 \implies V_A - 16.022 = 0$$

$$\mathbf{V_A = 16.02\text{ kN} \quad (\uparrow)}$$

Equilibrium Calculation Checklist

Step	Action	Common Pitfall to Avoid
1	Draw Free Body Diagram with all unknown support reactions	Forgetting H_A at pin supports or misidentifying normal direction on inclined rollers.
2	Convert UDLs / UVLs / Trapezoidal loads into equivalent point loads	Placing the equivalent load at the middle of the whole beam instead of the middle of the load span.
3	Resolve inclined forces into x and y components	Swapping $\sin \theta$ and $\cos \theta$ relative to the given reference axis.
4	Apply $\Sigma M_{\text{Hinge}} = 0$ to find one vertical support reaction	Including pure couple moments multiplied by distance. (Pure moments do not get multiplied by distance!).
5	Apply $\Sigma F_y = 0$ to find the remaining vertical reaction	Forgetting to include vertical components of inclined forces.
6	Apply $\Sigma F_x = 0$ to find horizontal hinge reaction	Omitting horizontal components of inclined roller reactions or loads.